

D2 — Solution architecture: data flow, product choices, custom components

Four layers plus checks plus render (D2 §1.1, condensed). Sources → **bronze** (verbatim, stamped with lineage) → **silver** (melted, crosswalk-resolved to one canonical observation grain) → **gold** (the marts a bulletin reads, plus two seeded statistical checks that always annotate, never gate) → publish to Parquet/DuckDB → render (Astro/Vega-Lite, static bulletin + email + Superset for the connected office). One Sand product used (Superset); the crosswalk, per-figure lineage, and the two checks are custom — the rest is standard warehouse plumbing, not the reason this took six weeks.



Red outline: Custom, built this cycle Dashed outline: Bought / adopted, scoped Grey outline: Sand product, not used in D2 Green outline: Delivered output

Synthetic test data throughout (assignment CSVs plus a hand-built DHIS2 sample) — not real Rwanda health records. Mart ERD and build-vs-buy detail in the full submission and repository.